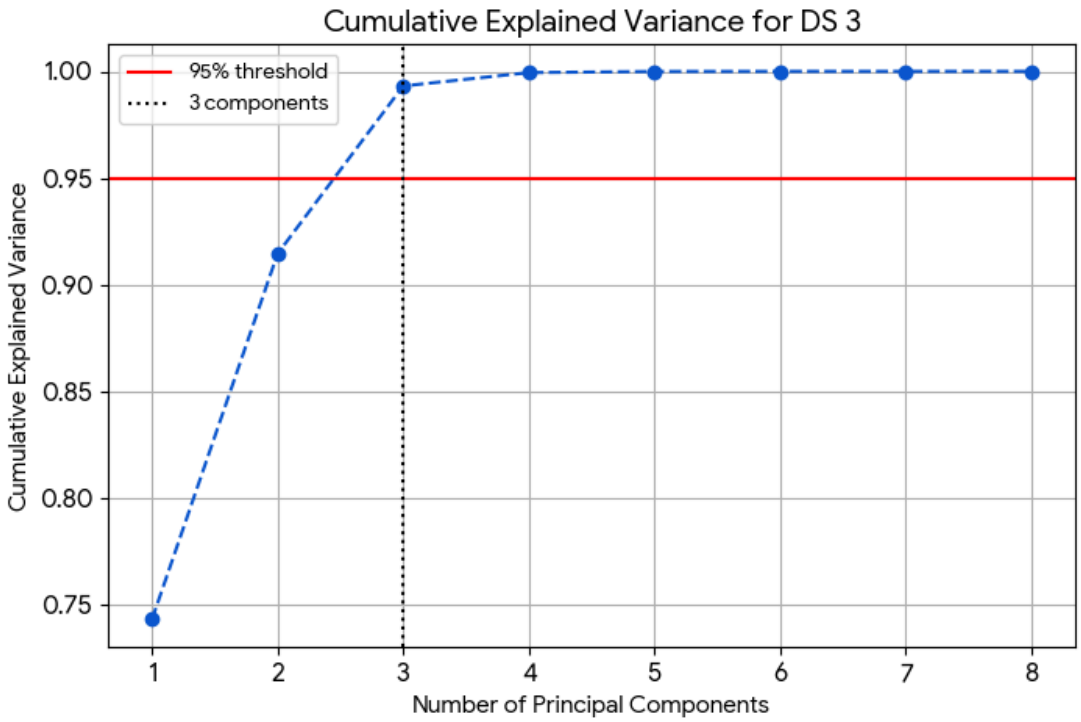
Assignment 2 (21d170001 - Aayush M Anand)

1. For the dataset, DS 3 , the analysis shows that 3 principal components are needed to capture over 95% of the variance in the data.

The plot below visualizes the result. We can see the "elbow" where the cumulative variance crosses the 95% threshold, which happens at the 3rd component.



2. Based on the analysis of the other datasets provided, here is the number of principal components required for each:

- DS 1: 1 principal component was needed.
- DS 2: 2 principal components were needed.
- DS 4: 4 principal components were needed

3. The results make a great deal of sense when viewed through the lens of basic power engineering principles. The number of principal components needed to describe the system is directly related to the number of *independent variables* or "degrees of freedom" in the data.

All datasets contain 8 variables that are physically linked by fundamental laws:

- **Ohm's Law:** Relates Voltage, Current, Resistance, and Reactance ($V=IZ$).
- **Power Equations:** Relate Active Power, Reactive Power, Apparent Power, Voltage, Current, and Power Factor ($S^2=P^2+Q^2$, $P=VI\cos\theta$, etc.).

Because of these strong relationships, the variables are highly correlated. PCA is excellent at identifying the underlying independent factors that cause variation across all these correlated variables.

My dataset DS 3 is similar to DS 2, but here the Resistance is held constant while the Reactance and Voltage are varied. This also introduces two main drivers of variation. The fact that it requires three components instead of two (like in DS 2) suggests that the interplay between voltage and a changing reactance introduces slightly more complex variance that requires an additional dimension to fully capture.